[SYNTHETIC TEST DATA / DEMONSTRATION FIXTURE - NOT OFFICIAL OEM SPECIFICATION]

AUTOSAR High-Level Design Document

Vehicle Control System - HLD Specification

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1. Introduction

1.1 Purpose and Scope

This High-Level Design (HLD) document defines the software architecture of the Vehicle Control System following the classic AUTOSAR software architecture standard.
Version 2.0 adds functional safety monitoring (SafetyManager) and updates signal precision specifications.

1.2 References and Standards

References include AUTOSAR Classic Platform Release 4.4.0, ISO 26262 ASIL-D Functional Safety Requirements, and corporate powertrain electronic control guidelines.

2. System Architecture Overview

2.1 Architecture Layers

The software architecture follows the standard three-tier AUTOSAR layer model consisting of the Application Layer containing Software Components (SWCs), the Runtime Environment (RTE) handling communication abstractions, and the Basic Software (BSW) managing low-level hardware abstraction and microcontroller drivers.

2.2 Component Overview

The system consists of five primary software components:
1. VehicleControl: Central application component computing motion control requests.
2. SensorManager: Acquires, filters, and broadcasts vehicle telemetry and dynamics.
3. DiagnosticManager: Manages DTCs and external diagnostic protocol services.
4. CommunicationManager: Manages gateway communication across CAN and V2X networks.
5. SafetyManager: Responsible for functional safety monitoring (ISO 26262 ASIL-D) and emergency override.

3. Software Components

3.1 VehicleControl

3.2 SensorManager

SensorManager is responsible for acquiring, filtering, and distributing sensor data from physical sensors. It produces VehicleSpeed as a float64 signal, EngineRPM as a uint16 signal, and EngineTemperature as a float32 signal through VehicleDataInterface.
Purpose: Sensor data acquisition and distribution.
Input Ports: RawSensorIn
Output Ports: VehicleSpeedOut, EngineRPMOut
Provided Interfaces: VehicleDataInterface
SensorManager produces VehicleSpeed which is consumed by VehicleControl.

3.3 DiagnosticManager

DiagnosticManager handles on-board diagnostic (OBD) operations, DTC management, and diagnostic service requests. It provides diagnostic information through DiagnosticInterface.
Purpose: Diagnostic services and DTC management.
Ports: DiagnosticRequestIn, DiagnosticResponseOut, DiagnosticStatusOut
Provided Interfaces: DiagnosticInterface
Dependencies: None (standalone diagnostic provider).

3.4 CommunicationManager

CommunicationManager manages external vehicle communication including CAN bus messages and V2X data exchange. It exchanges vehicle state information through V2XCommunicationInterface.
Purpose: External communication management.
Ports: CommDataIn, CommDataOut, VehicleStateIn
Used Interfaces: V2XCommunicationInterface
Dependencies: VehicleControl (receives state data).

3.5 SafetyManager

SafetyManager is a newly added component responsible for functional safety monitoring (ISO 26262). It monitors system integrity and can override control commands if safety violations are detected.
Ports: SafetyStatusOut, SystemHealthIn
Interfaces: SafetyMonitorInterface

4. Interfaces

4.1 VehicleDataInterface

Signal	Data Type	Unit	Producer	Consumer
VehicleSpeed	float64	km/h	SensorManager	VehicleControl
EngineRPM	uint16	rpm	SensorManager	VehicleControl
EngineTemperature	float32	degC	SensorManager	VehicleControl

4.2 DiagnosticInterface

4.3 V2XCommunicationInterface

Type: Sender-Receiver Interface
V2XCommunicationInterface handles V2X and internal vehicle state communication.
Signals: VehicleState, CommStatus

4.4 SafetyMonitorInterface

Type: Client-Server Interface
Signals: SafetyStatus, SystemHealth

5. Signals

Signal Name	Type	Unit	Source	Destination	Interface
VehicleSpeed	float64	km/h	SensorManager	VehicleControl	VehicleDataInterface
EngineRPM	uint16	rpm	SensorManager	VehicleControl	VehicleDataInterface
EngineTemperature	float32	degC	SensorManager	VehicleControl	VehicleDataInterface
VehicleState	enum	-	VehicleControl	CommunicationManager	V2XCommunicationInterface
DiagnosticStatus	uint8	-	DiagnosticManager	VehicleControl	DiagnosticInterface
ControlCommand	struct	-	VehicleControl	Actuators	-

6. Port Definitions

Port Name	Component	Direction	Interface	Signals
VehicleSpeedIn	VehicleControl	Input	VehicleDataInterface	VehicleSpeed
EngineRPMIn	VehicleControl	Input	VehicleDataInterface	EngineRPM
SafetyStatusIn	VehicleControl	Input	SafetyMonitorInterface	SafetyStatus
ControlCommandOut	VehicleControl	Output	-	ControlCommand
VehicleSpeedOut	SensorManager	Output	VehicleDataInterface	VehicleSpeed
DiagnosticRequestIn	DiagnosticManager	Input	DiagnosticInterface	-
DiagnosticResponseOut	DiagnosticManager	Output	DiagnosticInterface	-
SafetyStatusOut	SafetyManager	Output	SafetyMonitorInterface	SafetyStatus

7. Dependencies

7.1 Component Dependencies

VehicleControl depends on SensorManager for sensor data input.
VehicleControl depends on DiagnosticManager for system health monitoring.
VehicleControl depends on SafetyManager for safety override commands.
CommunicationManager depends on VehicleControl for vehicle state information.

Source	Depends On	Relationship	Description
VehicleControl	SensorManager	consumes data	Receives VehicleSpeed and EngineRPM
VehicleControl	DiagnosticManager	monitors health	Receives DiagnosticStatus
VehicleControl	SafetyManager	safety override	Receives SafetyStatus
CommunicationManager	VehicleControl	receives state	Gets vehicle state for V2X broadcast

8. Functional Flows

8.1 Sensor Data Flow

Physical sensors -> SensorManager -> VehicleDataInterface -> VehicleControl -> ControlCommand -> Actuators

8.2 Diagnostic Flow

External tester -> DiagnosticManager -> DiagnosticInterface -> VehicleControl